

**In one line:** GCD submits closures to *queues* and *blocks threads* to wait; **Swift Concurrency** *suspends tasks* at *await* (the thread is freed) and lets the **compiler** prove safety with **actors** (serialised state), `@MainActor` (UI) and `Sendable` (what may cross). Rule zero in both: **UIKit only on the main thread**.

## GCD

- DispatchQueue.main = **serial**, bound to the main thread (UI). DispatchQueue.global(qos:) = shared **concurrent** pool. DispatchQueue(label:) = serial unless attributes: .concurrent.
- async returns at once; sync **blocks the caller** until the block ran. main.sync from main (or sync onto the serial queue you are on) = **deadlock**.
- QoS**, high→low: .userInteractive > .userInitiated > .default > .utility > .background.
- DispatchGroup (enter/leave/notify) = wait for N jobs; DispatchSemaphore(value: N) = limit to N; .barrier on a *private concurrent* queue = reader-writer.
- URLSession completion handlers run on a background queue** (the session's delegate OperationQueue) — *never* main. Hop before UI: DispatchQueue.main.async {...}. Forgot .resume()? Tasks start suspended.

## Async/await

- await **suspends, never blocks**: the thread returns to a cooperative pool (one per core, no thread explosion); the task may resume on another thread.
- Task {} inherits actor, priority, task-locals. Task.detached {} inherits **nothing** (loses @MainActor). Both unstructured.
- Structured**: async let x = f() (fixed count), withThrowingTaskGroup (dynamic count). Parent waits for children; cancel flows down, errors flow up. Two plain awaits in a row run **one after the other**; async let a = f(); async let b = g() starts both at once.
- Cancellation is cooperative**: task.cancel() only sets a flag. Check Task.isCancelled or try Task.checkCancellation() (throws CancellationError).
- withCheckedThrowingContinuation bridges a callback API — resume **exactly once** (0 = hang, 2 = trap).
- actor**: reference type; one task at a time touches its state; outside calls need await. nonisolated opts a member out (methods that touch no isolated state; Sendable lets). **Reentrant**: at every await inside it, other calls may run ⇒ re-check state after await.
- @MainActor = global actor on the main thread; on a type, method or property. Off-main: await MainActor.run {}.
- Sendable = safe to cross isolation: value types of Sendable parts, final class with only let Sendable props, actors (implicitly). @unchecked Sendable = “I lock it myself”. Swift 6 makes violations *errors*.

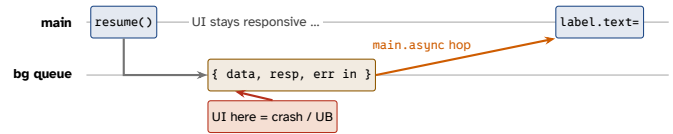
## Example — the main-thread hop

```
// 1. Completion handler: runs on a BACKGROUND queue
URLSession.shared.dataTask(with: url) { data, _, _ in
    let img = data.flatMap(UIImage.init(data:))
    DispatchQueue.main.async { self.imageView.image = img } // hop!
}.resume() // tasks start suspended

// 2. async/await inside @MainActor: resumes on main, no hop
@MainActor func load() async throws {
    let (data, _) = try await URLSession.shared.data(from: url)
    imageView.image = UIImage(data: data) // already on main
}
```

## Which thread runs what

### 1 · completion handler



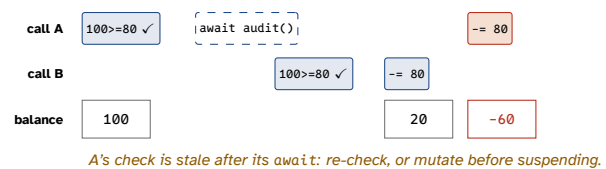
### 2 · await in @MainActor



### 3 · main.sync from main



## Actor reentrancy



## Traps

- Completion handlers run on URLSession's background queue** → hop to main for UI. async/await resumes on *your* actor — @MainActor needs no hop. A nonisolated async func leaves the actor — unless Swift 6.2's NonisolatedNonSendingByDefault (SE-0461) is on: then it runs on the caller's actor.
- await ≠ blocking**. sync/semaphore/sleep() block a thread; Task.sleep suspends.
- Task.detached in a @MainActor VM runs off-main**.
- Cancelling does not stop a loop that never checks**.
- DispatchGroup**: an early return without leave() → notify never fires; use defer.
- Actor as a “make it fast” tool**: it *serialises* — use a TaskGroup for parallel work.

## GCD → Swift concurrency

|                             |                                              |
|-----------------------------|----------------------------------------------|
| DispatchQueue.main.async    | @MainActor · await MainActor.run             |
| serial queue guarding state | actor                                        |
| DispatchGroup + notify      | async let · task group                       |
| semaphore “at most N”       | task group seeded with N, add one per finish |
| global().async              | Task.detached · @concurrent (6.2)            |
| asyncAfter                  | try await Task.sleep(for:)                   |
| callback API                | withCheckedThrowingContinuation              |

## Remember

**GCD blocks threads, await suspends tasks.** “Background brings the data, main paints it.” QoS: Interactive, Initiated, Default, Utility, Background — “I I D U B”.